

The hereditary-subalgebra semisimplicity question in arXiv:0907.3103 was answered by arXiv:1206.3466

Literature-status note

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Status: literature already answered. The later Theorem 4.2 proves the stronger exact formula for the Jacobson radical of every hereditary subalgebra.

1 The source question

For the source, an HSA (hereditary subalgebra) D of an operator algebra A is an approximately unital subalgebra satisfying $DAD \subset D$. Proposition 2.5 of arXiv:0907.3103 proves that semiprimeness and topological simplicity pass from A to its HSA's. Remark 2.6 immediately afterward, on PDF page 4, states:

We do not know if semisimplicity passes to HSA's.

Equivalently, if the Jacobson radical $J(A)$ vanishes, must $J(D)$ vanish for every HSA $D \subset A$? [1]

2 The exact later answer

Theorem 4.2 of arXiv:1206.3466 (PDF page 7) states that, for every HSA D in every operator algebra A ,

$$J(D) = D \cap J(A).$$

It follows immediately that $J(A) = 0$ implies $J(D) = 0$; hence semisimplicity does pass to HSA's [2].

The later result is strictly stronger than the yes/no assertion. Its proof uses the spectral characterization of the radical. One inclusion factors $x \in J(D)$ as $x = dy$ by Cohen factorization and uses $DAD \subset D$ to show $r(xa) = 0$ for every $a \in A^1$. The reverse inclusion uses the preceding spectral-invariance proposition: quasi-invertibility of elements of D agrees whether computed in D or in A .

The supporting paper introduces Theorem 4.2 with the explicit sentence “The following answers a question posed in [3],” and its reference [3] is the source paper. This is therefore an explicit later literature answer, not an agent-generated theorem. The other independent questions in arXiv:0907.3103 are outside this packet's scope.

References

- [1] M. Almus, D. P. Blecher, and S. Sharma, *Ideals and structure of operator algebras*, arXiv:0907.3103 (2009), Remark 2.6; J. Operator Theory 67 (2012), 397–436.

- [2] M. Almus, D. P. Blecher, and C. J. Read, *Ideals and hereditary subalgebras in operator algebras*, arXiv:1206.3466 (2012), Theorem 4.2; Studia Math. 212 (2012), 65–93.